Moller scattering 3

The Casimir trick uses matrix arithmetic to sum over spin states.

Let

$$
\begin{aligned}
f_{11} &= 8s^2 + 8u^2 - 64m^2 s - 64m^2 u + 192m^4 \\
f_{12} &= -8s^2 + 64m^2 s - 96m^4 \\
f_{22} &= 8s^2 + 8t^2 - 64m^2 s - 64m^2 t + 192m^4
\end{aligned}
$$

such that

$$
\begin{aligned}
\sum_{abcd} \mathcal{M}_1 \mathcal{M}_1^* &= \frac{\alpha^2}{t^2} f_{11} \\
\sum_{abcd} \mathcal{M}_1 \mathcal{M}_2^* &= \frac{\alpha^2}{tu} f_{12} \\
\sum_{abcd} \mathcal{M}_2 \mathcal{M}_2^* &= \frac{\alpha^2}{u^2} f_{22}
\end{aligned}
$$

By the Casimir trick we have

$$
\begin{aligned}
f_{11} &= \mathrm{Tr}\left((\not{p}_4 + m)\gamma^\mu(\not{p}_2 + m)\gamma^\nu\right) \mathrm{Tr}\left((\not{p}_3 + m)\gamma_\mu(\not{p}_1 + m)\gamma_\nu\right) \\
f_{12} &= \mathrm{Tr}\left((\not{p}_4 + m)\gamma^\mu(\not{p}_2 + m)\gamma^\nu(\not{p}_3 + m)\gamma_\mu(\not{p}_1 + m)\gamma_\nu\right) \\
f_{22} &= \mathrm{Tr}\left((\not{p}_3 + m)\gamma^\mu(\not{p}_2 + m)\gamma^\nu\right) \mathrm{Tr}\left((\not{p}_4 + m)\gamma_\mu(\not{p}_1 + m)\gamma_\nu\right)
\end{aligned}
$$

Recall that slash matrices are formed as

$$
\not{p} = p^\mu g_{\mu\nu} \gamma^\nu = p^0\gamma^0 - p^1\gamma^1 - p^2\gamma^2 - p^3\gamma^3
$$

Eigenmath script